

Revision Notes for Class 7 Maths Chapter 3 – A Peek Beyond the Point

Sometimes, when we need to measure objects very precisely, using standard whole number units like 1 cm or 1 meter is not enough. For example, when measuring the length of two similar-looking screws or the sharpness of a pencil, the measurement might fall between two whole numbers. In such cases, we divide the main unit into smaller pieces, like tenths, to get a more accurate answer. For instance, if a screw measures $2\frac{7}{10}$ cm, it means it is 2 centimeters plus 7 parts out of 10 small divisions between 2 cm and 3 cm.

The Need for Smaller Units

When we measure objects like a pen, sharpener, or small toy pieces, sometimes the answer will be a value between whole numbers, such as $2\frac{7}{10}$ cm or $3\frac{2}{10}$ cm. Dividing each centimeter into 10 equal parts lets us express more precise lengths. This division of units into tenths helps obtain measurements that are otherwise difficult if we are restricted to only whole numbers.

Understanding Tenths & Fractions

A number like $3\frac{4}{10}$ means 3 units and 4 out of 10 equal parts of the unit, also known as 3 point 4 (3.4) in decimal form. We can also write it as $\frac{34}{10}$, which represents thirty-four tenths. Ten tenths always make up one whole. These fractions help in understanding parts of a whole in daily life, especially while comparing or measuring different objects accurately.

How to Read and Write Fractional Units

When we see $2\frac{1}{10}$, we read it as “two and one-tenth”. Similarly, $\frac{4}{10}$ is read as “four-tenths” or “four one-tenths”. This helps in proper communication and understanding of decimals and fractions. For example, $4\frac{8}{10}$ units can also be written as $\frac{48}{10}$ units, just like $13\frac{1}{10}$ means thirteen and one-tenth.

Comparing and Adding Tenths

When we add two fractions like $2\frac{7}{10}$ and $3\frac{6}{10}$, we first add the whole parts ($2 + 3 = 5$) and the tenths ($7 + 6 = 13$). Since 13 tenths are more than one whole, we convert ten tenths into 1 unit and add the remaining 3 tenths. So, the answer becomes $6\frac{3}{10}$. The same is true for adding other values, or for arranging them in order according to their size. Fractions and decimals help compare the size or measurement of different items easily.

Place Value System and Decimal Point

Our Indian place value system is extended for decimals as well. The right side of the decimal point represents the tenths, hundredths, thousandths, and so on. For example, in 4.23, 4 is the whole number part, 2 is the tenth, and 3 is the hundredth part. The decimal point (.) separates the whole number from its fractional part so we can immediately recognize which part is less than a whole.

Operations with Decimals

Addition and subtraction with decimals is similar to whole numbers, but we must line up the decimal points carefully. For example, adding 3.25 and 2.6, we write:

$$\begin{array}{r} 3.25 \\ +2.60 \\ \hline =5.85 \end{array}$$

Keeping the decimal points in line ensures correct calculation. When subtracting, make sure both numbers have the same number of decimal places by adding zeros as needed.

Comparing Decimals

To compare two decimal numbers, first compare their whole parts. If the whole parts are equal, look at the tenths, then hundredths, and so on. For instance, between 3.4 and 3.41, 3.41 is bigger because, after the tenths place, the hundredths place has a "1" compared to zero.

Placing Decimals on a Number Line

Decimals are placed on a number line just as we place whole numbers — but between two numbers, we find marks for tenths, hundredths, and so on. For example, between 2 and 3 on a line, the point 2.7 would lie closer to 3, as it is seven-tenths past 2. This helps visualize and understand their relative values.

Key Properties of Decimals and Fractions

- A unit can always be split into 10 tenths, 100 hundredths, or 1000 thousandths depending on need.
- 10 one-hundredths make 1 tenth, and 100 one-hundredths make 1 unit.
- The decimal point is the separator for the whole and the part less than one.
- To add or subtract decimals, always align the decimal points.
- Comparing decimals is done starting from the leftmost (largest) place value.

Summary Table: Place Value (Decimals)

Whole Number Side	Decimal Side
Ten, One	Tenth, Hundredth, Thousandth
10 (whole)	0.1 (one-tenth), 0.01 (one-hundredth), 0.001 (one-thousandth)

The skills learnt in this chapter help us measure small objects, compare values using decimals and fractions, and carry out addition and subtraction of numbers with decimals. Remember to carefully align decimals and understand the place value of every digit while working with decimal numbers.